



Construction of a predictive model of shovel productivity applying machine learning algorithms

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Abstract

Currently, many mining companies face the need to increase the efficiency of their performance in conditions of growing competitiveness and market globalization. To achieve the set objectives, it is essential to reduce costs, optimize the use of equipment and human resources. Mining companies usually implement technological process control systems where a large amount of updated information about their production processes is stored, normally using long-term accumulated information that consists mainly of statistical reports. Taking the above into account, the need arises to use this accumulated information to identify segments of technological processes that are less efficient and be able to improve productivity. To do this, new technologies can be used such as the use of Machine Learning (ML). The present work focuses on the creation of a predictive model of “shovel” productivity, where shovels are heavy load equipment, used for any activity that involves moving earth or rocks in large volumes, with a level of acceptable confidence, which unlike the traditional method, include other quantitative and qualitative variables. The research began with the general hypothesis that the productivity model developed could improve the prediction of blade productivity with greater accuracy using Machine Learning models. From the results we obtained that the model that best predicts the productivity of the blades is the “Random Forest Regressor” with an accuracy of 99.7% for the training and 91.1% for the testing DataSet; while the worst Machine Learning precision model was “Dummy Regressor”.

Keywords Machine learning · Random forest · Productivity · Training · Shovel

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Introduction

Currently, companies are looking for a way to optimize resources, generate value in the business through greater productivity of their operation teams. In the case of the mining sector, it is no exception since each of its activities are critical and the impact that may be generated in any of them will significantly affect the company (Zelinska 2020).

In mining, several of the activities with the greatest economic impact are the loading, hauling and stripping of mineral. It is there where production is determined and the greatest number of operating equipment is used. Obtaining a large amount of data from them that allows generating a series of reports for those interested in the operations and the management of said companies. However, these data have not yet been used in their entirety; from these data the importance value of each of the variables that act in the operation could be identified, and from this generate a model that allows predicting the productivities of their teams.

Recently, mining and technology have been complementary since the beginning of mining operations, starting from the inclusion of equipment for loading and hauling and currently being shown in the areas called “Mine Dispatch” or in the “Integrated Operations Centers”, in where thousands of data are stored from the different areas involved in the process from geology, through the mine as a plant and ending in the ports with the marketing of the concentrate.

The use of data recorded on the servers of companies in different sectors is allowing the discovery of new forms of business and improvements to the achievements already fulfilled. In large mining companies around the world, Artificial Intelligence is gaining more importance every year, involving everything from the automation of operational and computer processes, to creating models that allow predicting and simulating events to estimate future results and attack the causes that could compromise development.

It is currently known that Open Pit mines there are problems of high waiting time in loading and hauling operations and to achieve a reduction in said time, good productivity is required in working with shovels. The Open Pit Mining Unit of our study is in the Department of Moquegua-Perú and is located at an altitude of 3400 meters above sea level. Currently, there is no prediction model for blade productivity and we work under the classic calculation model, which only contains the variables of moved tonnage, loading time and hang time. Nor is it identified which is the most important variable in the calculation.

We now know that machine learning is a powerful tool to improve productivity in mining through more accurate predictions, resource optimization, predictive maintenance and real-time analysis, which translates into greater efficiency and cost reduction. From this, the main contribution of our research is to determine a shovel productivity model in a Peruvian Mine, based on the operating variables recorded in the Dispatch system using Machine Learning models.

Our work is divided as follows: In Section “[Literature review](#)” we mention the literature review. In Section “[Current problems](#)” we mention the current problems. In Section “[Proposed methodology for the study](#)” we mention the methods and materials. In Section “[Results of the applied methodology](#)” we indicate the results and finally in Section “[Conclusions](#)” we indicate the conclusions of our developed work.

Literature review

As research works on the topic we have: In Soofastaei (2020) it describes the key challenges facing the mining sector as it transforms into a digital industry and provides guidelines on how data analysis should be implemented, as well as the approaches and methods to improve decision making.

In Hyder et al. (2019) indicated that artificial intelligence, machine learning and autonomous technologies provide many economic benefits to the mining industry through cost reduction, efficiency and productivity improvement, reducing worker exposure to hazardous conditions, continued production and improved safety. However, the implementation of these technologies has faced economic, financial, technological, labor and social challenges.

Then in Qi (2020) they mentioned that the Mining Industry faces great challenges such as the decrease in metal prices and the mineral grade. The world of opportunities is very small and Big Data is one of the most promising technological solutions.

Independently in Kang et al. (2023) indicated that, in the context of the globalization of the mining industry, evaluating the production viability of mining projects through smart technology is crucial to improve the efficiency of mining development. However, assessing the feasibility of such projects faces significant challenges due to incomplete data and complex variables. In recent years, the development of Big Data technology has offered new possibilities to quickly evaluate mining projects. From this, the authors perform an intelligent gold mine evaluation based on global mineral resource data to estimate whether a gold mine project can be put into production. The evaluation model they used was established based on the Random Forest model to quantitatively predict the viability of the mining project being brought into production.

In Yeganejou et al. (2022) mentioned that estimating the productivity of a truck fleet is one of the critical parts of a mining plan. In their work, the authors present a Monte-Carlo simulation model to imitate real truck and shovel operations, to measure the productivity of trucks. Independently, in Park and Choi (2013) presented a case study that carried out simulations on mineral transportation systems with shovel trucks in open pit mines in Korea. To do this, they used the GPSS/H simulation language to develop the simulation programs.

In Wu et al. (2017) mentioned that manufacturers present the need to develop predictive models that forecast mechanical failures and the remaining useful life (RUL) of manufacturing systems or components. Classical model-based or physics-based forecasts often require a deep physical understanding of the system of interest to develop closed-form mathematical models. The authors mention that with the rapid advancement of artificial intelligence, various Machine Learning algorithms have been developed and widely applied in many engineering fields. The objective of their research was to present a Random Forest (RF) based forecasting method for tool wear prediction.

Then in Choi et al. (2021) use different Machine Learning algorithms to predict mineral production in open pit mines, based on a truck transportation system supported by the Internet of Things (IoT). The results obtained by the authors

confirmed the viability and stability of the models to predict mineral production. Independently (Czaplicki 2008; Ercelebi and Bascetin 2009) mentioned that truck-shovel systems are the most used in open pit operations. In Fadin et al. (2017) mentioned that truck-shovel systems contribute 50 % of operating costs in open pit mines.

Independently in Ramezani and Tafazoli (2020) mentioned that incorporating intelligence into existing mining excavators improves mining safety, productivity, and energy efficiency. This can provide perception, monitoring and control capabilities that produce accurate and actionable data for mines. To monitor the smart excavator, there is an in-cab monitor that provides real-time status updates and guidance to operators, as well as a remote monitoring portal. Multiple sensors are integrated, including a rugged camera that observes the excavator bucket, high-resolution surveillance cameras, radar, hydraulic pressure monitoring, and electric motor power measurement.

Then in Dey et al. (2022) mentioned that loading, transportation of minerals and waste are the key operations in an open pit coal mine and therefore lead to a high operating cost, where the productivity of the mine can be increased by reducing the equipment loading cycle time and for this they used dump trucks optimally. To achieve this, the authors carry out a study of blade cycle times applying multiple regression and artificial neural network techniques.

Independently in Dindarloo and Siامي-Irdemoosa (2017) mentioned that downtime of mining equipment is costly and can be avoided with good intelligent and efficient maintenance planning. Modern mines today have large sets of data on equipment performance and reliability, from monitoring and dispatch systems. In their study the authors explore the application of classification and clustering approaches for the recognition and forecasting of failures in mining shovels. For clustering they used the k-means clustering algorithm and to predict future failures they used the support vector machine classification technique. Finally, the authors mentioned that these Machine Learning techniques are a valuable contribution to decision-making during their preventive maintenance program.

Independently in Ristovski et al. (2017) mentioned that open pit mining operations require the use of extremely expensive equipment, such as large trucks, shovels and loaders. To remain competitive, mining companies are under pressure to increase equipment utilization and reduce operating costs. The key to this in mining operations is to have sophisticated truck allocation strategies that ensure equipment is used efficiently with minimal operating cost. To address this problem, the authors have implemented a truck allocation approach that integrates machine learning, linear/integer programming, and simulation. The results

presented in their research demonstrate that machine learning can bring significant value to the mining industry.

In Shariati et al. (2019) presented a complete monitoring for excavators (smart excavator) based on neural networks to produce accurate and actionable data for a mine. The solution obtained by the authors allows increasing the efficiency of the blades, reducing unexpected downtime costs and allowing planned maintenance. In their work the authors use a convolutional LSTM-based object detection approach. Independently in Baek and Choi (2020) they predicted ore production using deep neural networks (DNN) for truck transportation systems in open pit mines. The proposed method used two DNN models that are designed to predict ore production during the morning and afternoon hauling sessions, respectively. In their results, the authors obtained good prediction accuracy.

Then in Aqueveque et al. (2021) present the development of an intelligent and easy-to-implement IoT monitoring system that uses vibration measurement devices to evaluate the real-time status of excavators, electric shovels and backhoes, in non-stationary operations in the mining industry. In their work, the authors mention that the total failure rates due to loss per fleet of machines can reach up to 30 %. When conducting field tests with bulldozers and backhoes from different manufacturers, machine learning algorithms were used that were able to classify the health status of the machine with an accuracy of between 85 % and 95 %.

In Patil et al. (2021) mentioned that their study aims to discuss the application of analysis to reduce machine downtime by predicting equipment failures, thereby improving the asset utilization of the fleet used in the mining industry. In their work, the authors used data obtained from the IoT, built predictive models for component failures (engine, brakes and transmission) and predictive models for the failure time of equipment for underground mining trucks and loaders, using supervised automatic learning algorithms, such as decision trees.

Then in Harmse et al. (2022) mentioned that the deep mining industry is experiencing increasingly tight profit margins due to increasing operating costs and declining production. The mining industry is known for its lack of dynamic control in complex embedded systems running at great depths, making it difficult to implement technologies that use IoT. An important integrated system in a typical underground mine is the cooling-ventilation system. The authors in their study develop and compare several Machine Learning prediction techniques to predict the integrated behavior of a key component that operates at the boundary of the cooling-ventilation system.

In Garouani et al. (2022) mentioned that the resources of industrial systems are capable of producing large amounts of

data. This data is typically in heterogeneous and distributed formats, but provides a means to extract information that can enable the deployment of intelligent management tools for production activities. Based on this, the authors present a software platform that can allow non-experts, who are not familiar with Artificial Intelligence (AI), to select and configure algorithms optimally adapted to their needs.

Independently in Chimunhu et al. (2022) mentioned that production planning and scheduling optimization for underground mining operations have continued to attract significant attention over the past decades. In their work the authors mention that several studies have been carried out to use mathematical programming models, such as mixed integer programming, heuristics and simulation algorithms, including combinations of these techniques for the optimization of production scheduling, with some notable achievements. In their work they also mention that the use of data analysis and machine learning can improve the effectiveness of planning and scheduling in underground mining.

Current problems

In mining, one of the activities with the greatest economic impact is the loading and transport of ore and waste rock. Production is determined in this area and the largest number of operating equipment is used. From these operations, such enriching data can be obtained that will allow a series of reports to be generated for those interested in the operations and especially for company management.

However, this data still presents complexity in the information. From this data, the importance value of each of the variables that act on the operation could be identified, and a model can be generated that allows us to predict the productivity of its equipment, through the use of machine learning models.

The Fig. 1 shows some equipment used in loading and transport, such as: shovels and trucks.

In our case study, there are problems of high hang time (time in which the shovels do not have trucks and are waiting for them), compromising the productivity of the shovels. In order to solve this problem, initially we will analyze each of the variables that intervene in loading and hauling.

To do this, we made an ISHIKAWA Diagram (Ishikawa and Loftus 1990), in which we will show the causes that compromise the main problem, which is the productivity of the blades, as shown in Fig. 2. In which we could see in detail the variables that influence more on the increase in production (Availability of equipment, both shovels and trucks, the month in which mining is taking place since the dry season and the rainy season are not the same, the guard and shift since the operating conditions of day are better due to the visualization of the terrain, the EFH which is the equivalent horizontal hauling distance and the operating parameters) and leads us to ask ourselves the following questions.

Can a mathematical model of blade productivity be determined with the operating variables recorded in the Dispatch system using a Machine Learning model? Will this model be the one with the highest degree of prediction than other Machine Learning models to determine blade productivity? What are the most important and influential variables in blade productivity?

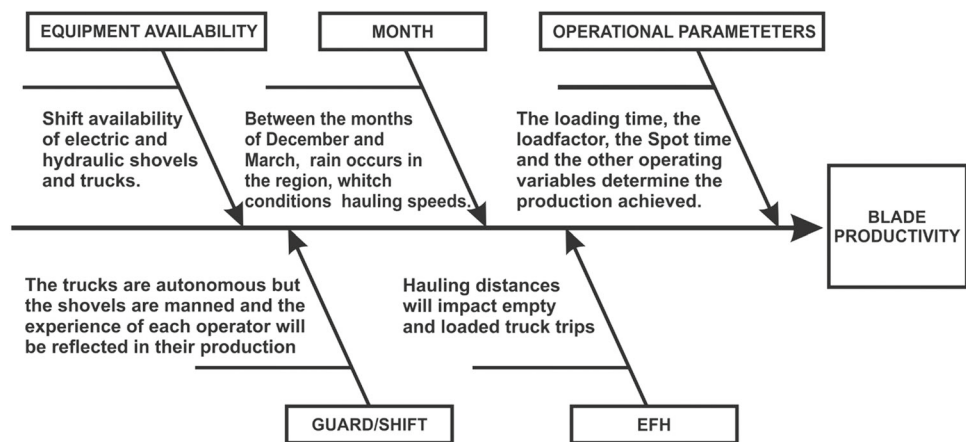
From the ISHIKAWA diagram of the Loading and Hauling Process Fig. 1. It is concluded that you can work on five aspects.

- Availability of equipment: The availability of the loading and hauling fleet delivered by the maintenance area is one of the main conditions of the production that can be achieved by each shovel.
- Guard / shift: The human factor (operators' skill) and working conditions (day or night) significantly affect

Fig. 1 Some equipment used in loading and transport



Fig. 2 ISHIKAWA diagram of the loading and hauling process in an open pit operation



production, reducing the visualization of roads and increasing safety measures to avoid accidents.

- **Month:** In the Open mining unit under study, between the months of December and March, there are times of moderate to intense rain, impacting the safety conditions of the roads, reducing the speeds reached by trucks and increasing hauling time, as well as the restriction of some areas of the mine due to slope stability.
- **EFH (Equivalent horizontal hauling distance):** When presenting an inverse relationship with respect to the production of a blade, it is important to determine the degree of influence in a way that allows decisions to be made in situ and obtain better results.
- **Operational parameters:** Among the important parameters we have:
 - **Loading time:** It is the time that transports the material to its destination.
 - **Spot time:** It is the positioning time of the trucks to be loaded.
 - **Queue (%):** This refers to the relationship between the time that a truck or hauling equipment spends waiting (in line) to be loaded by a shovel or excavator, and the total time of the hauling cycle.
 - **Hang (%):** Reflects the proportion of time the shovel or truck is not performing productive tasks, such as loading or transporting material.
 - **Average empty trip time:** Is the average travel time that a truck makes to a shovel after unloading.

- **Average loaded trip time:** This refers to the average time it takes a shovel or truck to move from the loading site to the unloading site.
- **Unloading time:** it is the time the truck needs to deposit the material at the destination.

All these parameters are reflected in the use of the equipment and the production that can be achieved.

Figure 3 shows the main operations in an open pit mine.

The open-pit mining operation includes several operations such as: drilling and blasting, loading, hauling and transportation, crushing and screening, where shovels and excavators are usually the first equipment used in the material handling process.

It is necessary to mention that the productivity of shovels in surface mining can be affected by three key parameters: loading time, positioning time (spot) and waiting time (hang). Among these, hang time is especially critical when hauling equipment availability is low, since longer truck return times delay the loading cycle, thus reducing shovel efficiency. Furthermore, if the loading equipment does not operate at 100% of its mechanical capacity, loading times increase, which also negatively impacts productivity. It is essential to keep shovels in optimal operating conditions to avoid these delays.

On the other hand, it is important to note that maintenance stops and delays of the shovel itself do not directly affect productivity, since the monitoring system records these events and adjusts the operating time according to the status of the

Fig. 3 Open pit mine operations
Ramezani and Tafazoli (2020)

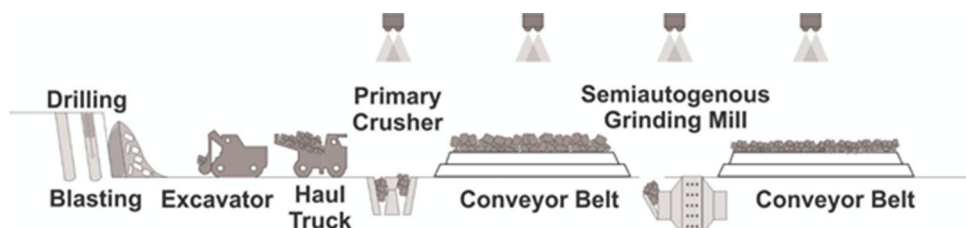
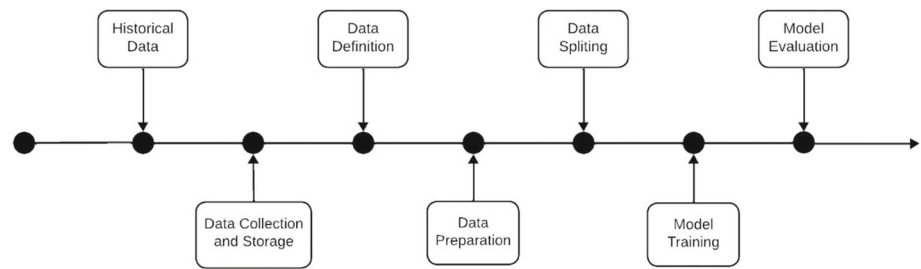


Fig. 4 Methodology used

equipment, preventing them from being counted as lost productive time.

Proposed methodology for the study

To validate our proposal, we will use the following methodology, which consists of several steps that are applied to obtain our results. This is shown in Fig. 4.

- **Data Collection and Storage**
Data obtained from loading and hauling operations are stored in the Mine Dispatch system.
- **Data Definition**
The next step to build our model is to define the input and output data.
- **Data Preparation**
Consists of data preparation, null value analysis and data cleaning with normalization.
- **Data Splitting**
This stage consists of dividing the data of our model for training and testing.
- **Model Training**
Using various supervised Machine Learning models, such as: Random Forest (RF), Decision Trees (DT), Linear Regression (LR), Gradient Boosting (GB), Dummy Regressor (DR), AdaBost Regressor (AR), Bayesian Ridge (BR) among others.
- **Model Evaluation**
Performance metrics are obtained for each model in learning such as: MAE, MSE, RMSE and R^2 . Then it is compared which model has the best metric in predicting the output variable.

To achieve our optimal design, various Python 3.0 libraries were used such as: Scikit-learn, Pandas, Matplotlib and Seaborn, TensorFlow, Keras, PyCaret. The tests were carried out on the computer in the University laboratory, whose specifications are shown below: Processor: Intel Core i9, Hard drive: 7 TB, RAM: 128 GB, Graphic Card (GPU): NVIDIA RTX 3090 with 24 GB of VRAM.

Data collection

The data will be extracted from the Dispatch System servers of the study Mine, through the use of inquiries with Querys in SQL Server, through the connection to the company's Virtual Private Network (VPN).

The Query inquiry extracts data from dimension tables, which are primary or characterization tables of the study variables with their respective codes, and from the fact tables, which is the production table that records each of the trips made by all trucks and the production of each shovel in operation.

The period of our analysis runs from December 31, 2021 to March 8, 2023. The extracted records were 1,096,270 between truck trips and shovel loading records, which were organized into production records by year, month, shift, day, time, fleet of shovels and trucks, creating a Dataset of 28,170 final records, with which the study was carried out.

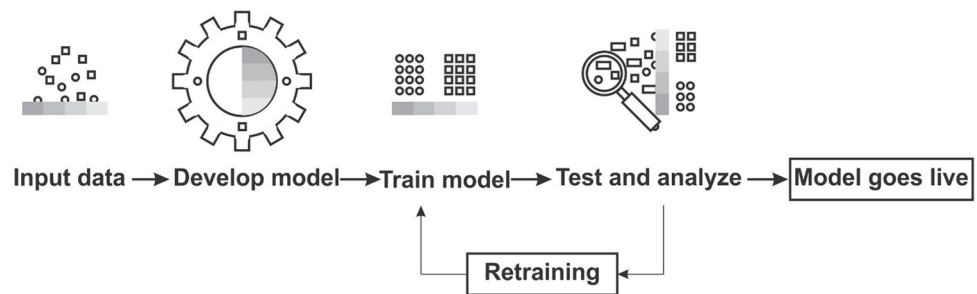
Variable selection

Our Dataset has 31 variables, which are the following: Year, month, date, shift, guard, time, primary fleet, secondary fleet, secondary equipment, material, type of material, origin phase, destination phase, tonnage, queue time, spot time, loading time, loaded trip time, queue time at destination, spot time at destination, unloading time, empty trip time, hang time, HT availability, SH availability, load factor, empty EFH, loaded EFH and cycle. The dependent variable being: Productivity.

Learning techniques of machine learning for processing

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that focuses on the use of data and algorithms to mimic the way humans learn, with gradual improvement in accuracy. According to Saravanan and Sujatha (2018), the intention of ML is to allow computers to learn without human assistance.

Fig. 5 How the machine learning process works SAP (2023)



Using statistical methods, algorithms are trained to make classifications or predictions, and discover key information within data mining projects. This key information facilitates decision making within applications and businesses, ideally impacting growth metrics. Figure 5 shows the operation of the Machine Learning process.

The processing carried out by Machine Learning consists of separating the Dataset into two subgroups, the first being the training Dataset in which the algorithm will be applied to determine the predictive model; and a second group that will be the testing Dataset in which tests will be carried out with the inputs determining the target variable to verify the reliability of the determined model.

Learning methods of machine learning

There are different types of learning methods that we mention below.

- **Supervised**

It uses labeled data sets, training algorithms to accurately classify data or predict outcomes. As input data is entered into the model, it adapts its weights until it has been properly fitted.

This occurs as part of the cross-validation process to ensure that the model avoids overfitting or underfitting. Supervised learning enables organizations to solve a wide variety of real-world problems at scale. Some methods used in supervised learning are: Neural Networks, Naïve Bayes, Linear Regression, Logistic Regression, Random Forest and the support vector machine (SVM).

- **Unsupervised**

Analyze and group unlabeled data set clusters. These algorithms discover hidden data groupings or patterns without the need for any human intervention. This method's ability to discover similarities and differences in information makes it ideal for exploratory data analysis, cross-selling strategies, customer segmentation, and image and pattern recognition. It is also used to reduce the number of features in a model through the process of dimensionality reduction.

Principal component analysis (PCA) and singular value decomposition (SVD) are two of the most common

approaches to perform this process. Other algorithms used in unsupervised learning are neural networks, K-means clustering, and probabilistic clustering methods.

- **Semi-supervised**

It becomes a viable solution when there are large amounts of raw and unstructured data. This model consists of introducing small amounts of labeled data to increase the unlabeled Dataset. Essentially, labeled data acts to jump-start the system and can significantly improve the speed and accuracy of learning. A semi-supervised learning algorithm instructs the machine to analyze labeled data based on correlative properties that could apply to unlabeled data.

- **For reinforcement**

The machine receives the reference response and learns by finding correlations between all the correct results. The reinforcement learning model does not include a baseline response, but rather introduces a set of allowed actions, rules, and potential end states. When the desired goal of the algorithm is fixed or binary, machines can learn by example. But in cases where the desired outcome is mutable, the system must learn by experience and reward. In reinforcement learning models, the “reward” is numerical and is programmed into the algorithm as something the system seeks to collect.

Results of the applied methodology

Identification of variables

Our Dataset contains 31 variables among which is the dependent variable “Productivity”. There are 28170 travel records, some of them being numeric (integers, decimals) and objects (text), as evidenced in Table 1.

Null detection

Using the “isnull()” function from the Pandas library, we calculate that the number of null values is 0 for each variable.

Table 1 Dataset information

Number	Column	Non-null Count	Dtype
0	Year	28170 non-null	int64
1	Month	28170 non-null	int64
2	Day	28170 non-null	object
3	Shift	28170 non-null	object
4	Guard	28170 non-null	object
5	Hour	28170 non-null	int64
6	SecondaryMachineClassName	28170 non-null	object
7	SecondaryMachineName	28170 non-null	object
8	PrimaryMachineClassName	28170 non-null	object
9	Material	28170 non-null	object
10	Material Type	28170 non-null	object
11	Origin phase	28170 non-null	object
12	Destination phase	28170 non-null	object
13	Tonnage	28170 non-null	float64
14	Queue time	28170 non-null	float64
15	Spot time	28170 non-null	float64
16	Charging time	28170 non-null	float64
17	Loaded travel time	28170 non-null	float64
18	Destination queue time	28170 non-null	float64
19	Destination spot time	28170 non-null	float64
20	Unloading time	28170 non-null	float64
21	Empty travel time	28170 non-null	float64
22	Load factor	28170 non-null	float64
23	Loaded EFH	28170 non-null	float64
24	Empty EFH	28170 non-null	float64
25	HT % Availability	28170 non-null	float64
26	Hang time	28170 non-null	float64
27	SH % Availability	28170 non-null	float64
28	Ton	28170 non-null	float64
29	Cycle	28170 non-null	float64
30	Productivity	28170 non-null	int64

Descriptive statistical analysis of the variables

Next, we performed an initial descriptive analysis of each of the numerical variables, finding their mean, standard deviation, minimum value, maximum value and quartiles, which are detailed in Table 2. For a better understanding, each of the variables will be explained in detail:

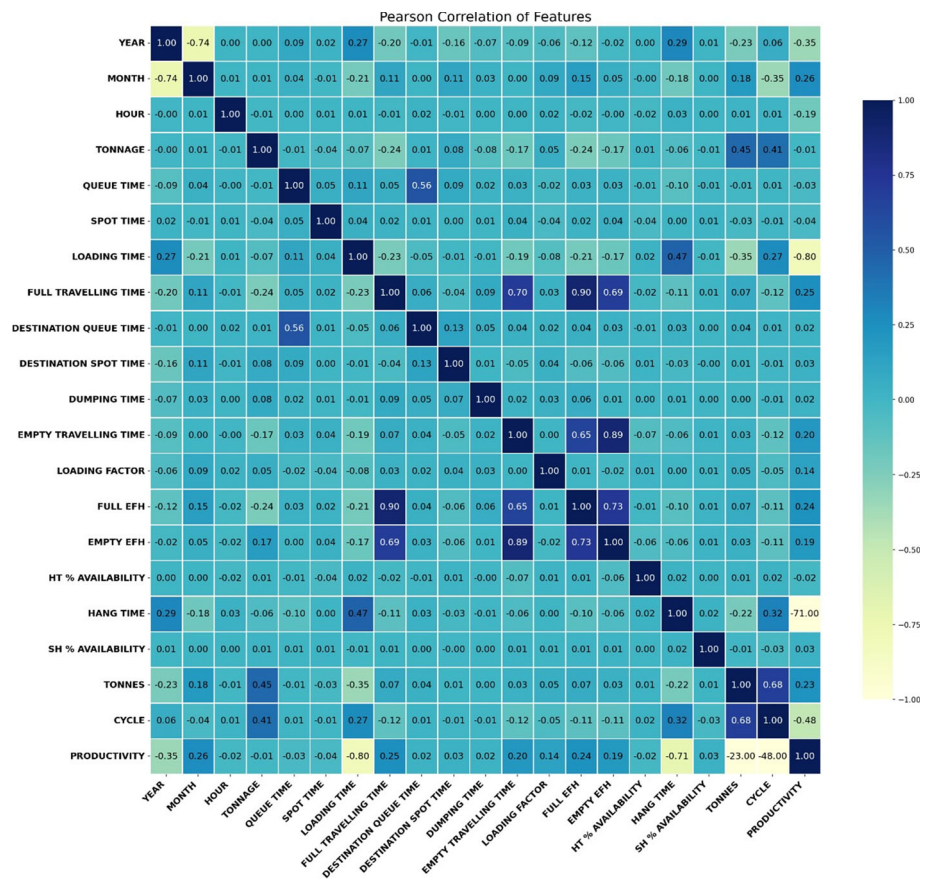
- Queue time: is the waiting in line time of the trucks at the shovel.
- Spot time: it is the positioning time of the trucks to be loaded.
- Charging time: it is the time it takes to be charged.
- Loaded travel time: it is the time that transports the material to its destination.
- Destination queue time: it is the queue of trucks to unload at the destination.
- Destination spot time: it is the positioning time of the trucks to unload.
- Unloading time: it is the time the truck needs to deposit the material at the destination.
- Empty travel time: is the travel time that a truck makes to a shovel after unloading.
- Load factor: it is the average tonnage that a truck loads per shovel.
- Loaded EFH: is the equivalent horizontal distance that the truck travels loaded.
- Empty EFH: is the equivalent horizontal distance traveled by the empty truck.
- HTs Availability: is the availability of the trucks to be used.
- Hang time: it is the time that the shovel does not have trucks and has to wait for them.

Table 2 Descriptive analysis of the numerical variables

Variables	Media	Std	Min	25%	50%	75%	Max
Queue time (min)	2.95	140.44	0.00	1.38	2.49	3.95	28.08
Spot time(min)	0.69	16.01	0.00	0.57	0.64	0.75	9.68
Charging time (min)	1.93	53.14	0.02	1.20	1.55	2.60	10.07
Loaded travel time (min)	9.55	276.42	0.00	6.00	8.79	12.51	50.32
Destination queue time (min)	0.82	80.32	0.00	0.00	0.31	1.10	27.20
Destination spot time (min)	0.77	47.67	0.00	0.54	0.61	0.68	23.97
Unloading time(min)	0.83	34.75	0.00	0.67	0.74	0.83	24.25
Empty travel time (min)	8.21	243.88	0.00	4.96	7.58	10.88	42.27
Load factor (t)	310.10	14.09	3.30	303.70	311.38	318.30	368.40
Loaded EFH (km)	3.86	1897.56	0.00	2.45	3.79	4.95	35.43
Empty EFH (km)	3.88	2098.34	0.00	2.28	3.67	4.93	24.38
HTs availability (%)	98.98	6.62	3.82	100.00	100.00	100.00	100.00
Hang time (min)	1.84	59.80	0.00	1.18	1.52	2.15	6.81
SH availability (%)	99.94	1.10	61.22	100.00	100.00	100.00	100.00
Productivity (t/h)	4781.68	1819.42	2000.06	2946.81	5197.44	6045.19	10769.43

- SHs Availability: is the availability of the blades.
- Productivity: it is the tonnage per hour that a shovel can move.

To understand the relationship between the variables, we graph our correlation matrix as shown in Fig. 6. In which we observe that the correlation is low between all the variables, thereby confirming that the Machine Learning model

Fig. 6 Correlation between the dataset variables

for “Linear Regression” is completely ruled out. Therefore, another prediction model was sought in the research work.

Outlier detection and data cleaning

Outliers are atypical values and are an observation that is numerically very different from the rest of the elements in a sample. These data can cause us problems in interpreting what happens in the processes carried out in the mine. In Mensi et al. (2022) mentioned that in recent years, a wide variety of outlier detectors have been proposed in the literature.

Then it is recommended to clean the outliers before generating the Random Forest Regressor model, to obtain a higher degree of precision, the operation criterion must be taken into account to select the correct data and not lose important information, this cleaning does not must exceed 5% of the total data. For example, we generate Fig. 7 as shown in the box plot.

It is observed that the tonnages produced by the CAT 7495 shovels (SH001, SH002, SH003) are greater than 9000 tons/hr, which are considered outliers since they rarely exceed this limit, given the current conditions of the mining operation.

To maintain homogeneity, we eliminated the outliers of material moved per hour, not affecting the amount of data, but helping to have a more coherent model. This is shown in Fig. 8.

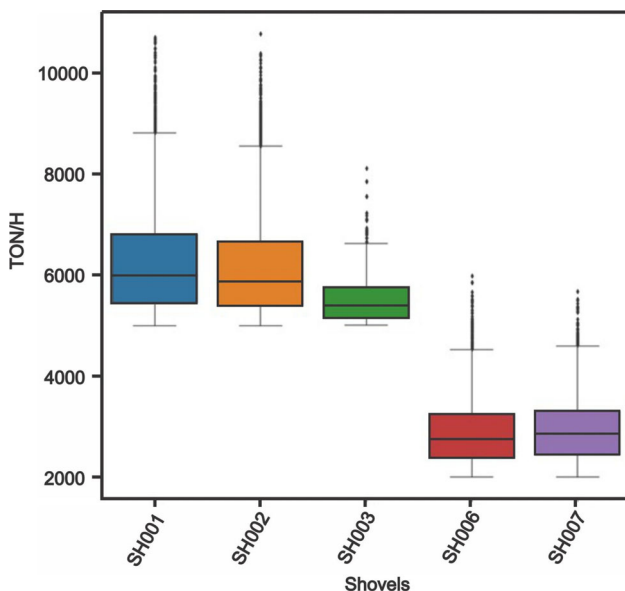


Fig. 7 Box plot for tons produced by each shovel

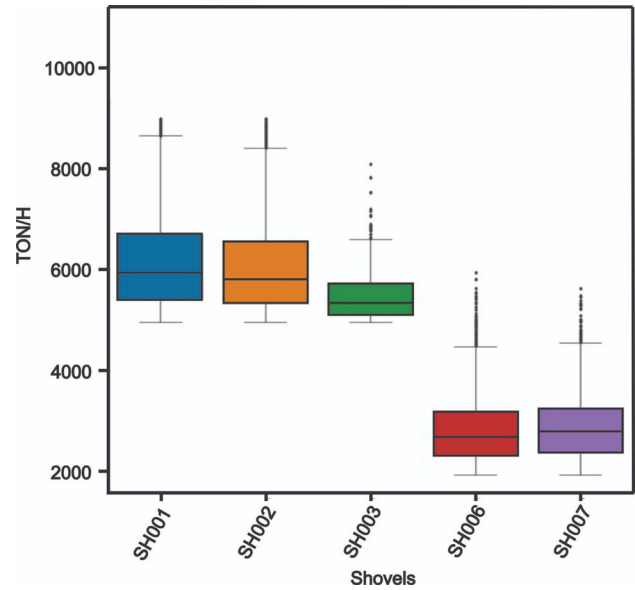


Fig. 8 Box plot of tonnage produced per hour without considering outliers

Preparation of the dataset to train

Before modeling, we must begin by converting the categorical variables into numerical ones with the help of the dictionary that was designed as partially shown in Table 3.

Table 3 Dictionary of categorical variables

Variables	Value	Variable code
Turn	Day	1
Turn	Night	2
SecondaryMachineClassName	CAT 7495	2
SecondaryMachineClassName	CAT 6060BH	3
SecondaryMachineName	SH001	3
SecondaryMachineName	SH002	4
SecondaryMachineName	SH003	5
SecondaryMachineName	SH005	6
SecondaryMachineName	SH007	7
PrimaryMachineClassName	CAT 794AC CMD	2
Material	WC	1
Material	WA	2
Material	SSA	3
Material	SSB	4
Material	WV	5
Material	W0	6
Material	HYB	7
Material Type	Clearing	1
Material Type	Ore	2

Table 3 continued

Variables	Value	Variable code
Origin phase	F01N	1
Origin phase	T05S	2
Origin phase	F01S	3
Origin phase	F02N	4
Origin phase	Adhoc	5
Origin phase	B01A	6
Origin phase	T02D	7
Origin phase	T02N	8
Origin phase	S01N	9
Origin phase	B01O	10
Origin phase	T04N	11
Origin phase	T06S	12
Origin phase	F03S	13
Origin phase	S01S	14
Origin phase	F03B	15

Next, we select the final DataSet that will enter the modeling, which will only have numerical variables (eliminating the categorical ones and leaving those that were parameterized).

We generate an internal ordering of the columns leaving the target variable “Productivity” at the end of the DataSet, with which we would obtain the following columns: Year, Month, Date, Shift, Guard, time, Secondarymachineclass-name (SMCN), Secondarymachinename (SMN), Primarymachineclassname (PMCN), material, type of material, phase origin, destination phase, tonnage, queue time, spot time, loading time, loaded travel time, queue time at destination, spot time at destination, unloading time, empty travel time, load factor, loaded EFH, empty EFH, HTS availability (%), hang time, SH availability (%), Shift Cod, SH Fleet Cod, SH Cod, Material Cod, Material Type Cod, Origin Phase

Cod, Destination Cod, Productivity. Table 4 shows the partial Dataset that enters our model.

Evaluation and choice of the best machine learning model

Next, we will evaluate the various Machine Learning models used to obtain the best predictive model of blade productivity for our case study.

It is important to have metrics that allow us to compare the effectiveness of the models, applying Machine Learning models, obtaining for each of them the following prediction error rates such as: Mean Square Error (MSE), Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), as shown in Table 5.

From Table 5 we observe:

- **Ensemble Models**

Random Forest Regressor and Gradient Boosting Regressor are the best models in this table, with the lowest errors (MAE, MSE, RMSE) and the highest R^2 values. These models excel in scenarios where there are complex and non-linear relationships in the data.

Adaboost Regressor shows lower performance than the other two ensemble models, but is still competitive, although with a higher error and a lower R^2 than the previous ones.

- **Linear and Regular Models**

Linear Regression, Ridge Regression, Lasso Regression, Elastic Net, and Lasso Least Angle Regression (LARS) show intermediate performance. These models assume a linear relationship between the variables, which can be limiting in more complex problems. Still, they are useful for their simplicity and interpretability.

Bayesian Ridge also follows a similar performance line, without offering significant advantages over the other linear models.

Table 4 Partial Dataset to be entered into the modeling

Index	Date	Shift	Guard	Hour	SMCN	SMN	PMCN
7	2022/06/01	Day	2	9	CAT 6060BH	SH006	CAT 794AC CMD
8	2022/06/01	Day	2	9	CAT 6060BH	SH006	CAT 794AC CMD
9	2022/06/01	Day	2	10	CAT 6060BH	SH006	CAT 794AC CMD
23	2022/06/01	Day	2	14	CAT 6060BH	SH006	CAT 794AC CMD
45	2022/06/01	Night	3	0	CAT 6060BH	SH006	CAT 794AC CMD
.....							
80888	2023/08/09	Day	1	15	CAT 7495	SH002	CAT 794AC CMD
80893	2023/08/09	Day	1	16	CAT 6060BH	SH007	CAT 794AC CMD
80894	2023/08/09	Day	1	16	CAT 6060BH	SH007	CAT 794AC CMD
80897	2023/08/09	Day	1	16	CAT 7495	SH002	CAT 794AC CMD
80898	2023/08/09	Day	1	16	CAT 7495	SH002	CAT 794AC CMD

Table 5 Classification of predictive models of blade productivity with Machine Learning for the analyzed data

Model	MAE	MSE	RMSE	R^2 (%)
Random Forest Regressor	65.580	9285.040	96.36	99.71
Gradient Boosting Regressor	261.66	146795.09	382.99	95.59
Decision Tree Regressor	223.69	208518.54	455.69	93.74
Linear Regression	445.57	368561.81	606.75	88.94
Ridge Regression	445.55	368549.82	606.74	88.94
Bayesian Ridge	445.17	368507.75	606.70	88.94
Lasso Least Angle Regression	445.22	369011.26	607.11	88.93
Lasso Regression	445.22	369011.31	607.11	88.93
AdaBoost Regressor	505.02	389041.24	623.47	88.31
Elastic Net	451.71	401576.60	633.36	87.95
Huber Regressor	434.77	535946.42	731.37	83.92
K Neighbors Regressor	453.88	548404.85	739.95	83.54
Orthogonal Matching Pursuit	488.36	594954.44	770.86	82.15
Passive Aggressive Regressor	572.85	598053.91	767.79	82.09
Dummy Regressor	1593.32	3330630.48	1824.88	-0.03

• Alternative Models

Huber Regressor is a robust model against outliers, but in this case it does not provide a significant improvement compared to standard linear models.

K Neighbors Regressor (KNN) and Orthogonal Matching Pursuit (OMP) show lower performance, with higher errors and lower R^2 . These models may be less effective in scenarios where complex or non-linear relationships need to be captured.

- Reference Models Dummy Regressor acts as a baseline reference, simply predicting the mean of the target values. As expected, it has the worst performance, highlighting the effectiveness of more advanced models.

In conclusion, in this analysis we found the superiority of the ensemble models on this dataset obtained, especially the Random Forest Regressor and Gradient Boosting Regressor. So for our case study, the model with the highest prediction was the Random Forest and the one with the lowest prediction was the Dummy Regressor.

Calculation of the best hyperparameters of the model

Every Machine Learning model is fed with parameters or hyperparameters to analyze and train the information. The Random Forest Regressor is a meta-estimator that groups classification decision trees into different subclasses to use the average to improve predictive accuracy and monitor overfitting error. Then the size of the subclass with the max samples hyperparameter is important; because if it is not controlled, the data set will be used to design each decision tree.

The hyperparameters that we must take into account, according to the Scikit-learn library, will be:

- n estimators: is the number of trees that you want to implement in the modeling.
- criterion: is the error of a grouping, which shows the quality as it will be treated. These can be “squared error” if the error is mean squared, which is equal to the variance reduction for feature selection, “friedman mse”, If the error is mean squared with the Friedman score it allows for potential splits, “absolute error” if absolute mean, through the median for the terminal nodes, and “poisson” for the Poisson deviation. Training with “absolute error” is more cumbersome to process than “squared error”.
- max Depth: allows you to determine the maximum depth of the tree. If the None option is selected, nodes reach all sheets until they are pure or contain less than simple min samples split.
- min samples split: Indicates the minimum number of samples required to split an internal node. If int is chosen, min samples split is considered the minimum number; in case of choosing float, the min samples splites will be a fraction and is the minimum number of samples for each split.
- min samples leaf: It is the minimum number of samples required to be in a leaf node. It serves to help the effect of smoothing the model, even more so in regression.
- max features: It is the number of features that we must evaluate when searching for the best tree..
- max leaf nodes: helps relative reduction of impurity. If we leave it as None by default, an unlimited number of nodes in the tree leaves will be considered.

- random state: Manipulates the randomness of bootstrapping the trees as the sampling of features when searching for the best decision tree.

According to Indrasiri et al. (2021) after obtaining the accuracy of each model for each classifier, an adjustment of the hyperparameters is performed. From there, library functions such as RandomizedSearchCV are used to obtain the optimal functions.

Through the implementation of the hyperparameters: “score samples”, “predict”, “predict_proba”, “decision function”, “transform”, and “inverse transform”, methods are used that optimize the cross validation search on the determined configuration. Not all the parameter values are used, but rather the configurations are set to determine the values that acquire the parameters of the distributions. The variable that determines parameter configurations is the n iter.

When all parameters are presented in a list, sampling without replacement will be performed. Continuous distributions should be used for continuous parameters.

To calculate them, the following ranges were determined:

- test size = 0.25: that is, 25 % of the data will make up the testing group and the model will be trained with the remaining 75 %.
- Random state = 38
- n estimators = [200, 400, 600, 800, 1000, 1200, 1400, 1600, 1800, 2000]
- max features = ['auto', 'sqrt']
- maxdepth = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100, None]
- min samples leaf = [1, 2, 4]
- Bootstrap = [True, False]

Below we present the programming code to apply cross-validation and determine the best hyperparameters for our Random Forest Regressor model.

Python code fragment

```
from sklearn.model_selection import train_test_split
train_features, test_features, train_labels, test_labels =
    train_test_split(X, Y, test_size = 0.25, random_state =
    38)
```

```
from sklearn.ensemble import RandomForestRegressor
rf = RandomForestRegressor(random_state = 38)
# Checking the parameters used by the current forest
print('Parameters currently in use:\n')
print(rf.get_params())
```

```
from sklearn.model_selection import RandomizedSearchCV
import numpy as np

# Number of trees in the random forest
n_estimators = [int(x) for x in np.linspace(start = 200, stop =
    2000, num = 10)]
# Number of features considered at each split
max_features = ['auto', 'sqrt']
# Maximum levels of the trees
max_depth = [int(x) for x in np.linspace(10, 100, num = 10)]
max_depth.append(None)
# Minimum number of samples required to split a node
min_samples_split = [2, 5, 10]
# Minimum number of samples required at each leaf node
min_samples_leaf = [1, 2, 4]
# Method for selecting samples for training each tree
bootstrap = [True, False]
```

```
# Create the random grid
random_grid = {'n_estimators': n_estimators,
               'max_features': max_features,
               'max_depth': max_depth,
               'min_samples_split': min_samples_split,
               'min_samples_leaf': min_samples_leaf,
               'bootstrap': bootstrap}

print(random_grid)
```

```
# Use the random grid to search for best hyperparameters
# First create the base model to tune
rf = RandomForestRegressor()

# Random search of parameters, using 3 fold cross validation,
# search across 100 different combinations, and use all
# available cores
rf_random = RandomizedSearchCV(estimator = rf,
                               param_distributions = random_grid,
                               n_iter = 100, scoring='
                               neg_mean_absolute_error',
                               cv = 3, verbose=2, random_state
                               =38, n_jobs = -1)

# Fit the random search model
rf_random.fit(train_features, train_labels)

# Fitting 3 folds for each of 100 candidates, totalling 300
fits
```

After processing the cross validation, we were able to obtain these parameters whose values are reflected in Table 6.

Table 6 Best hyperparameters for the Random Forest Regressor model

Hyperparameter	Best value
Test size	0.25
Random state	38
n estimators	2000
min samples split	4
min samples leaf	2
max features	'sqrt'
max depth	100

Table 7 Comparison of the predicted values and those recorded in the training

Model Random Forest	R^2 training	R^2 testing	MAE	MSE	RMSE
Value	99.71	91.10	65.58	9285.04	96.36

Application of the random forest regressor model

Once the best parameters were obtained, these were entered into our Random Forest Regressor model as seen in part of the program code.

```
from sklearn.model_selection import train_test_split
train_features, test_features, train_labels, test_labels =
    train_test_split(X, Y, test_size = 0.25, random_state =
    38)

# Training and testing the random forest
from sklearn.ensemble import RandomForestRegressor
rf_reg2 = RandomForestRegressor(n_estimators=2000,
                               min_samples_split=4,
                               min_samples_leaf=2,
                               max_features='sqrt',
                               max_depth=100,
                               bootstrap=False)

regressor = rf_reg2.fit(train_features, train_labels)
y_pred = regressor.predict(test_features)
```

Then in Table 7 the results obtained are shown.

Evidence

According to Uçar et al. (2020), mentioned that there is a proportion of the data between training and testing according to the total data. For our study to train our model in Random Forest Regressor, the DataSet was divided into 25 % for testing and 75 % for training.

To validate our hypothesis, it is verified in Table 8 and the data obtained from the first 10 values returned by the model and their respective records from the training DataSet are compared. In the same way, Table 9 is obtained, which shows the comparison of the values predicted in testing.

Independently, Fig. 9 shows the prediction error using the Random Forest Regressor algorithm.

Figure 9 shows that the R^2 is 0.911, which means that the model explains 91.1% of the variance in the data. This is a

Table 8 Comparison of the predicted values and those recorded in the training

Y	1	2	3	4	5	6	7
Y pred	5966.2	6352.6	5488.3	2629.5	6161.9	5751.8	5246.4
Y train	5951.9	6480.5	5470.1	2587.5	6265.8	5736.8	5189.0

Table 9 Comparison of predicted and recorded values in the testing

Y	1	2	3	4	5	6	7
Y pred	5524.2	7276.4	3620.8	2872.8	6229.5	5872.1	7259.1
Y test	5434.1	8584.3	2209.0	2583.0	6061.1	5626.7	8441.4

good indicator that the model is working correctly and that its predictions are fairly aligned with the actual values.

Then in Fig. 10, the predicted productivity and the productivity assigned in the test DataSet are shown.

Finally, by applying “feature importances” from the pandas library, the degree of importance of each of the variables that make up the model was obtained, the main ones being: Cod SH Fleet, Cod SH, Loading time and Hang time as seen in Fig. 11.

The classic model currently used does not allow quantifying the importance or influence of the variables in the model; while the built model determines the importance for each variable as shown in Table 10.

Discussion of results

There are several works related to solving some mining operations with the use of Machine Learning (ML) such as: In Baek and Choi (2020) they proposed a method based on Deep Neural Networks (DNN) to predict mineral production by truck transport systems in open-pit mines. The method proposed by the authors uses two DNN models that are designed to predict mineral production during the morning and afternoon transport sessions, respectively. As a result, they found

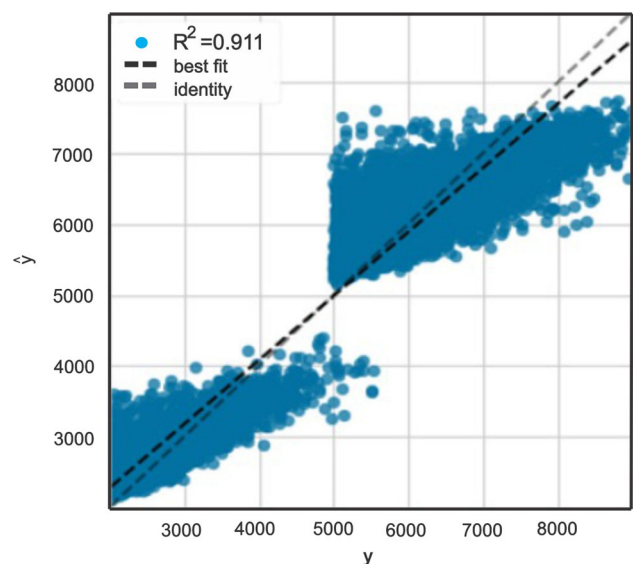
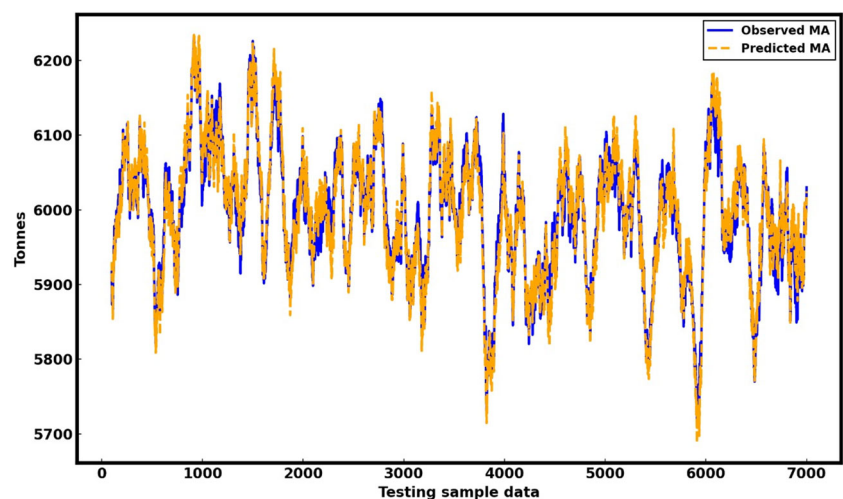
**Fig. 9** Prediction error in the testing DataSet applying Random Forest Regressor

Fig. 10 Predicted vs assigned productivity in the testing DataSet



that the R^2 value in both cases was 0.99, which leads to a high accuracy of mineral prediction.

Then, in Choi et al. (2021) they evaluated the feasibility of using different Machine Learning algorithms to predict mineral production in an open-pit mine in South Korea based on a truck transportation system supported by the Internet of Things (IoT). In their work, the authors used 6 ML algorithms such as: the Random Forest (RF), the Support Vector Machine (SVM), Multilayer Perceptron Neural Networks (MLP neural networks), the classification and Regression Tree, the k Nearest Neighbors and the M5Tree model. The results obtained by the authors revealed that the models used can potentially be used to predict mineral production in open-pit mines. The SVM, MLP and RF models were the ones that presented high precision, and within this, the SVM model showed the highest precision with an $R^2=0.988$.

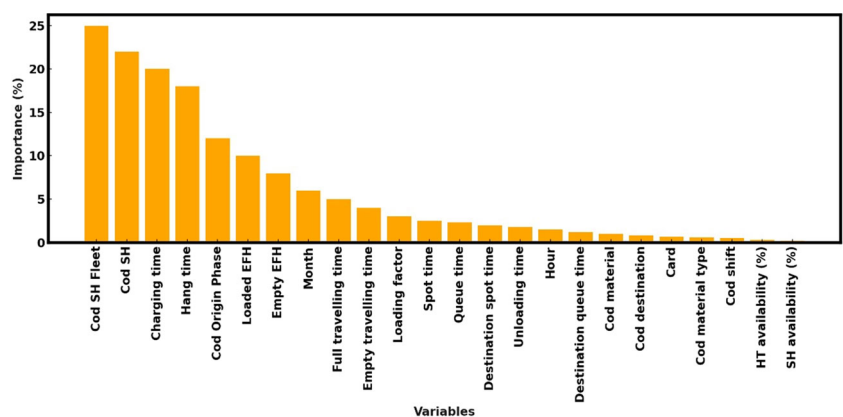
Independently in the study by Nobahar et al. (2022) they presented a Machine Learning (ML) method to optimize the loader type and the number of trucks used to meet the mineral demands of a processing plant in a kaolin mine in Zenouz (Iran). In their work the authors used 5 algorithms such as: Gradient Boosting (GB), Random Forest Regresoor

(RF), k-Nearest Neighbor (KNN), Linear Regression (LR) and Decision Tree Regressor (DT). Finding that the Gradient Boosting (GB) algorithm obtained an $R^2=0.85$ accuracy.

In Cotrina Teatino et al. (2024) they applied Machine Learning techniques to predict the production of an open-pit mine in Peru. The authors used 4 ML algorithms such as: Random Forest (RF), Extreme Gradient Boosting (XGBoost), K-Nearest Neighbors (KNN) and Bayesian Regression (RB). The results obtained indicated that the Bayesian Regression model was the most effective in predicting production in the open-pit mine with an $R^2=0.47$.

Then in Kumah et al. (2024) they indicated that it is necessary to develop reliable models to accurately predict mining production to improve operational efficiency. The authors in their work used 4 machine learning algorithms such as: Artificial Neural Network (ANN), Random Forest (RF), Gradient Boosting Regression (GBR) and Decision Tree (DT) to predict mining production. As a result, they found that in general the algorithms used manage to predict mining production with good accuracy, with the ANN being the one with the best $R^2=0.80$.

Fig. 11 Most important variables of our model



Although our work is oriented to another mining operation such as loading and applied to the prediction of shovel productivity in a copper and molybdenum concentrate production mine, we believe that it is difficult to compare. However, from 15 algorithms used in our work we found that the best model was the Random Forest Regressor with an $R^2=0.911$, which indicates that it has a high accuracy in predicting shovel productivity.

Some recommendations should be taken into account, such as:

- Before starting each Machine Learning model, you must look for the best hyperparameters of the model, which you want to develop with the help of the “Cross Validation” method, since it will allow for better prediction.
- The greater the amount of data in the DataSet, the better the prediction model. If, despite this, the model’s prediction is low, the less important variables must be eliminated, since these affect the interaction of the decision trees.

Table 10 Importance of the variables in the Random Forest Regressor Model

Variable	Importance (%)
Cod SH Fleet	24.9259%
Cod SH	23.2086%
Charging time	19.3239%
Hang time	16.0613%
Cod Origin Phase	2.7542%
Loaded EFH	1.9929%
Empty EFH	1.8462%
Month	1.3963%
Full travelling time	1.3190%
Empty travelling time	1.0892%
Loading factor	0.8684%
Spot time	0.7272%
Queue time	0.7172%
Destination spot time	0.6224%
Unloading time	0.5976%
Hour	0.5431%
Destination queue time	0.5123%
Cod material	0.5084%
Cod destination	0.3555%
Guard	0.2771%
Cod material type	0.1874%
Cod shift	0.1111%
HT availability(%)	0.0408%
SH availability(%)	0.0140%

- To corroborate the prediction of the model, tests must be carried out with the training data and also identify the level of prediction of this model with the data in the testing, since, if the training data reaches a high degree of prediction, but in the testing is low, the main cause is because overfitting is generated, that is, it became accustomed to training, but its degree of prediction is really not high. Therefore, it is recommended to also evaluate the prediction of the model with the testing data, which must also be good above 80%.

Conclusions

For the Machine Learning model, all the variables present in the Dataset were used, including the qualitative variables, which were categorized before training.

Of the Machine Learning models, as seen in Table 5, the model that best predicts the productivity of the blades is the “Random Forest Regressor” with an accuracy of 99.7 % for the training and 91.1% for the testing DataSet; while the worst Machine Learning model is “Dummy Regressor” with -0.03 % accuracy.

Classic shovel productivity models consider only the tons moved, the loading time and the hang time determined on a shovel. While the proposed Random Forest Regressor model takes into account the mentioned variables and adds the queue time, the spot time, the loaded travel time, the queue time at the destination, the spot time at the destination, the unloading time, empty travel time, load factor, SH fleet code, SH code, source phase code, destination phase code, loaded EFH, empty EFH, month, time, material, the type of material, the guard, the availability of trucks and shovels. All this information is shown in Table 10, which also shows the importance of each variable in the model.

As future work, we hope to test our model, but only using the most important variables such as: Code SH fleet (Secondarymachineclassname), Cod SH (Secondarymachinenename), Charging time and Hang time, which were determined in our work as shown in Fig. 11.

Author Contributions V.J. and J.R. made substantial contributions to the design of the work. J.R. and J.C. contributed to the acquisition and analysis of the data. V.J. prepared all figures. J.R. prepared all the tables. J.C. and V.J. revised it critically for important intellectual content. All authors reviewed and approved the version to be published.

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Data Availability No datasets were generated or analysed during the current study.

Code Availability Not applicable.

Declarations

Ethics approval This article does not contain any studies with human participants performed by any of the authors.

Competing interests The authors declare no competing interests.

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